

NEXT LEVEL TECHNOLOGIES

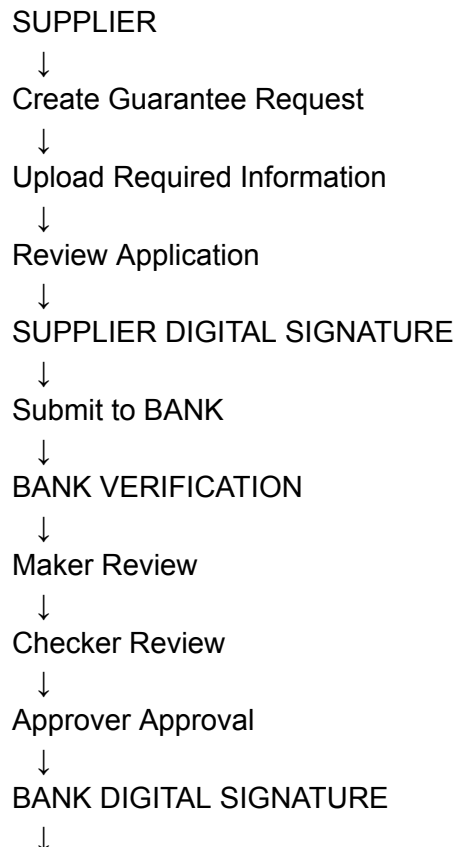
ADVANCED DIGITAL GUARANTEE PLATFORM

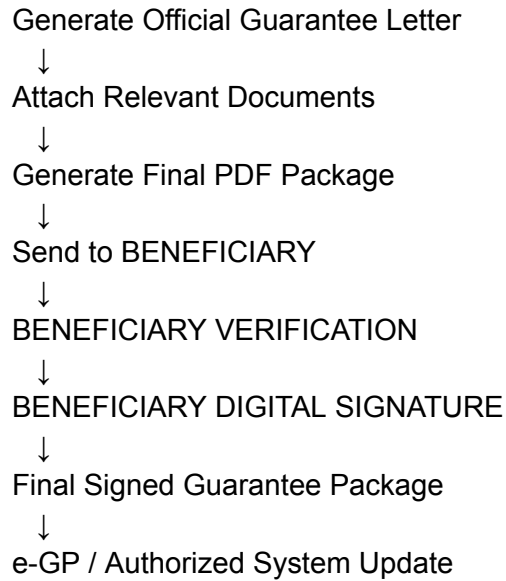
Revised Digital Signature & Guarantee Delivery Workflow

1. Core Principle

The platform must treat the guarantee as a **digitally signed, traceable document lifecycle**, not simply as a database status.

The primary flow is:





2. SUPPLIER PORTAL — DIGITAL SIGNATURE

Screen: Guarantee Request

Supplier completes:

- Guarantee Type
- Tender ID / Contract ID
- Beneficiary
- Amount
- Currency
- Validity
- Bank
- Purpose
- Required information
- Supporting documents

Before submission:

Application Summary

Guarantee Type	Performance Guarantee
Amount	ETB 10,000,000
Beneficiary	ABC Authority

Validity 365 Days
Bank XYZ Bank

Documents 8/8 Complete
Application READY

CTA:

Review & Sign

3. SUPPLIER DIGITAL SIGNATURE SCREEN

This is mandatory before the request reaches the bank.

DIGITAL SIGNATURE	
Guarantee Request: GR-2027-000123	
I confirm that the information and documents submitted in this application are accurate and authorized by me / my organization.	
Signature Certificate: ✓ Available	
[Sign Digitally]	

After signing:

✓ SIGNATURE SUCCESSFUL

Signed By:
ABC Construction PLC

Certificate:
Valid

Timestamp:
2027-01-10 10:42:21

Document Hash:
XXXXXXXXXXXXXX

Status:
DIGITALLY SIGNED

The signed request becomes a new immutable/versioned document.

4. SEND TO BANK

Only after supplier signature:

SUPPLIER SIGNED



VALIDATE SIGNATURE



CREATE BANK REQUEST



ENCRYPT / SECURE TRANSMISSION



BANK PORTAL / BANK API

Bank receives:

Guarantee Request Package

- Request ID
- Supplier information
- Tender/Contract information
- Guarantee type
- Amount
- Beneficiary
- Validity
- Supplier-signed application
- Supporting documents
- Digital signature metadata

Status:

SENT TO BANK

5. BANK PORTAL — INCOMING REQUEST

Bank Maker sees:

NEW GUARANTEE REQUEST

Request ID: GR-2027-000123

Supplier:

ABC Construction PLC

Guarantee:

Performance Guarantee

Amount:

ETB 10,000,000

Beneficiary:

ABC Authority

Supplier Signature:

✓ VALID

Documents:

8/8

Customer:

✓ VERIFIED

Actions:

Start Review

Request More Information

Reject

6. BANK VERIFICATION

Bank system performs:

Customer

- Customer identification
- KYC
- Account
- Customer status

Financial

- Guarantee limit
- Current exposure
- Available limit
- Collateral
- Fees

Digital Signature

- Certificate validity
- Signer identity
- Signature integrity
- Timestamp
- Document hash

Result:

CUSTOMER	✓
ACCOUNT	✓
LIMIT	✓
COLLATERAL	✓
DOCUMENTS	✓
SUPPLIER SIGNATURE	✓

7. BANK MAKER → CHECKER → APPROVER

Maker

↓

Checker

↓

Approver

Each person signs/approves according to the bank's configured authorization policy.

No user should be able to approve their own maker transaction where segregation-of-duties rules prohibit it.

8. BANK GUARANTEE LETTER GENERATION

After approval, the bank generates the official guarantee.

The system should not send only raw data to the beneficiary.

It creates an **Official Digital Guarantee Package**.

Main Guarantee Letter

Contains:

- Bank name/logo
 - Guarantee number
 - Date
 - Applicant/Supplier
 - Beneficiary
 - Guarantee type
 - Guarantee amount
 - Currency
 - Effective date
 - Expiry date
 - Terms and conditions
 - Claim conditions
 - Authorized bank information
 - Digital signature
 - Verification information
 - QR code
-

9. RELEVANT INFORMATION ATTACHMENTS

The bank can attach/configure relevant documents such as:

- Guarantee letter
- Supplier application
- Contract reference
- Supporting documents
- Bank approval information
- Terms and conditions
- Amendments
- Other authorized attachments

The platform must maintain an **Attachment Manifest**.

Example:

DIGITAL GUARANTEE PACKAGE

01	Official Guarantee Letter	✓
02	Supplier Application	✓
03	Contract Reference	✓
04	Supporting Document	✓
05	Bank Terms	✓
06	Amendment	-

10. FINAL PDF GENERATION

This is a major UX requirement.

After bank approval:

Generate Final PDF Package

The system creates a professionally formatted PDF containing:

PAGE 1

Official Guarantee Letter

PAGE 2

Guarantee Terms & Conditions

PAGE 3+

Authorized Supporting Documents

The PDF must have:

- Unique Guarantee ID
- Version number
- QR code
- Digital signature information
- Document hash / verification reference
- Issue date
- Expiry date
- Issuing bank
- Beneficiary
- Supplier

Actions:

Preview PDF

Download PDF

Send to Beneficiary

11. PDF VERSION CONTROL

Never overwrite signed documents.

Example:

Guarantee
NG-2027-00123

Version 1.0
Original Issuance

Version 1.1
Amendment

Version 1.2

Extension

Each version retains its own:

- PDF
 - signature
 - timestamp
 - hash
 - approval history
 - audit trail
-

12. BANK DIGITAL SIGNATURE

Before delivery to the beneficiary:

Generate Guarantee



Bank Review



Bank Approval



Generate Final PDF



BANK DIGITAL SIGNATURE



Signature Validation



Publish Guarantee

The bank's authorized signatory/signatories digitally sign the official guarantee document according to the bank's policy.

Result:

BANK SIGNATURE: VALID ✓

13. BENEFICIARY PORTAL — INCOMING GUARANTEE

Beneficiary receives notification:

New Digital Guarantee Received

Dashboard:

Guarantee ID
NG-2027-00123

Supplier
ABC Construction PLC

Bank
XYZ Bank

Amount
ETB 10,000,000

Status
PENDING BENEFICIARY ACCEPTANCE

Actions:

View

Verify

Download PDF

Accept

Reject

Request Clarification

14. BENEFICIARY VERIFICATION

Beneficiary opens the guarantee.

System automatically checks:

Bank Identity	✓	
Guarantee Number		✓
Digital Signature	✓	
QR Verification	✓	
Supplier	✓	
Beneficiary	✓	
Amount	✓	
Validity	✓	
Document Integrity		✓

Then:

GUARANTEE AUTHENTIC

15. BENEFICIARY PDF DOWNLOAD

Beneficiary must have:

Download Official PDF

Options:

- Download Guarantee Letter
- Download Full Package
- Download Signed Version
- Print

Downloaded document should visibly contain:

Digitally Signed

Verification QR

Guarantee ID

Document Version

16. BENEFICIARY ACCEPTANCE

This is a critical feature.

Beneficiary clicks:

Accept Guarantee

System displays:

ACCEPT DIGITAL GUARANTEE

I confirm that I have reviewed and accepted
this digital guarantee and the information
contained within it.

Guarantee:
NG-2027-00123

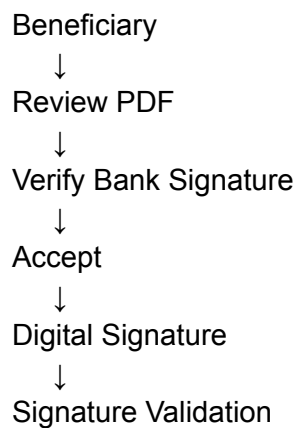
Amount:
ETB 10,000,000

[Digitally Sign & Accept]

Beneficiary then digitally signs.

17. BENEFICIARY DIGITAL SIGNATURE

Flow:



↓
Acceptance Recorded

Result:

✓ GUARANTEE ACCEPTED

Accepted By:
ABC Authority

Digital Signature:
VALID

Timestamp:
2027-01-10 14:32:10

Version:
1.0

18. BENEFICIARY REJECTION

Beneficiary can select:

Reject

But rejection must require a reason.

REJECT GUARANTEE

Reason:
[Select]

Comments:
[_____]

Supporting Document:
[Upload]

[Digitally Sign Rejection]

Possible configurable reasons:

- Incorrect amount

- Incorrect beneficiary
- Incorrect supplier
- Incorrect validity
- Incorrect guarantee type
- Missing information
- Invalid document
- Other

The rejection becomes an auditable event.

19. ACCEPTANCE PACKAGE

After beneficiary acceptance:

SUPPLIER SIGNATURE

✓

BANK SIGNATURE

✓

BENEFICIARY SIGNATURE

✓

↓

FULLY EXECUTED DIGITAL GUARANTEE

The final record contains all signatures.

20. GUARANTEE STATUS

Recommended lifecycle:

DRAFT

↓

SUPPLIER SIGNED

↓

SUBMITTED TO BANK

↓

BANK REVIEW

↓
BANK APPROVED
↓
BANK SIGNED
↓
ISSUED
↓
SENT TO BENEFICIARY
↓
PENDING ACCEPTANCE
↓
BENEFICIARY ACCEPTED
↓
ACTIVE

Alternative paths:

BANK REJECTED
BENEFICIARY REJECTED
CANCELLED
AMENDED
EXTENDED
CLAIMED
DISPUTED
RELEASE REQUESTED
RELEASED
FORFEITED
EXPIRED
CLOSED

21. CLAIM FLOW

Once the guarantee is **ACTIVE**:

Beneficiary:

Raise Claim

Claim Details

↓
Supporting Evidence
↓

Submit



Digital Signature



Bank Review

Bank:

Claim Review



Accept

OR

Reject

OR

Request More Information

OR

Dispute

Beneficiary portal:

Claim Status

Submitted

Under Review

More Information Required

Accepted

Rejected

Disputed

Settled

22. RELEASE FLOW

Beneficiary:

Request Release

Select Guarantee



Reason



Supporting Evidence



Review

↓
DIGITAL SIGN
↓
Submit to Bank

Bank:

Release Request
↓
Review
↓
Approve
↓
Release Collateral
↓
Close Guarantee

23. DISPUTE FLOW

If beneficiary disagrees with a bank/claim decision:

Raise Dispute

Dispute
↓
Reason
↓
Evidence
↓
Beneficiary Digital Signature
↓
Submit
↓
Bank Response
↓
Supplier Response where applicable
↓
Review
↓
Decision
↓
Resolved / Escalated

Every response can be digitally signed where legally/operationally required.

24. COMPLETE DIGITAL SIGNATURE CHAIN

The most important architecture is:



25. DIGITAL SIGNATURE RECORD

For every signature store:

Signature ID
Signer ID
Organization ID
Role
Certificate Reference
Document ID
Document Version
Document Hash
Signature Timestamp
Signature Status
Verification Result

Do not store private signing keys inside the application database.

The platform should integrate with the approved digital-signature/certificate infrastructure.

26. AUDIT TRAIL

Every critical action produces an event:

10:10 Supplier created request
10:14 Supplier uploaded documents
10:17 Supplier digitally signed
10:18 Request sent to Bank
10:30 Bank Maker reviewed
10:45 Checker approved
11:02 Approver approved
11:05 Bank digitally signed
11:06 PDF generated
11:07 Sent to Beneficiary
11:20 Beneficiary downloaded PDF
11:25 Beneficiary verified
11:30 Beneficiary digitally signed
11:31 Guarantee ACTIVE

27. NOTIFICATION FLOW

Supplier

- Submitted
- Bank received
- More information required
- Approved
- Issued
- Beneficiary accepted
- Rejected
- Claimed
- Released
- Disputed

Bank

- New request
- Supplier signed
- Approval pending
- Claim received
- Release received
- Dispute received

Beneficiary

- Guarantee received
- Verification completed
- Acceptance required
- Claim update
- Release update
- Dispute update

28. FRONTEND SCREEN ADDITIONS

The previous 3 portals should therefore include these additional screens:

Supplier

S-01 → S-18

Add:

- Review & Sign
- Digital Signature

- Signed Request
- Submission Receipt

Bank

B-01 → B-20

Add:

- Signature Validation
- Guarantee Letter Builder
- Attachment Manager
- PDF Preview
- PDF Generator
- Bank Digital Signature
- Delivery Center

Beneficiary

BE-01 → BE-18

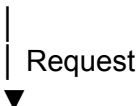
Add:

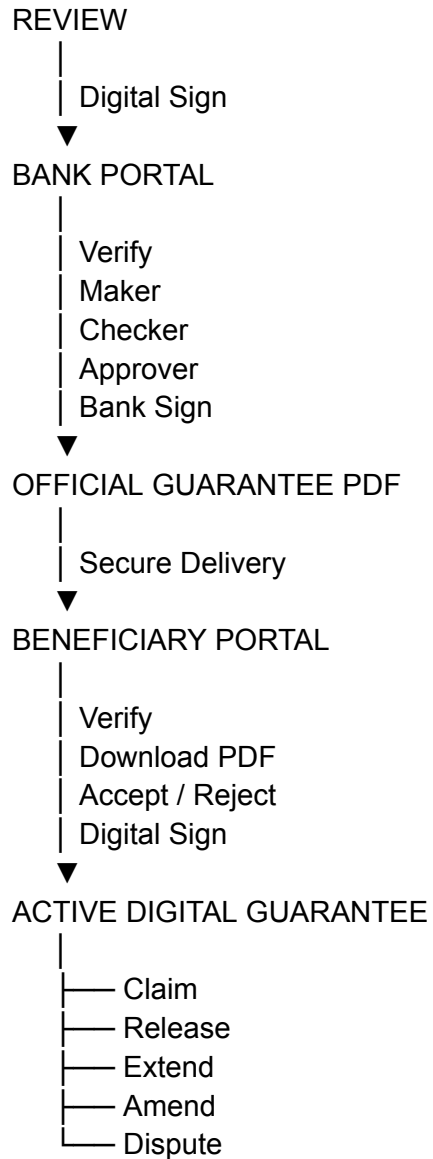
- Incoming Guarantee
- Verification
- PDF Viewer
- PDF Download
- Accept & Sign
- Reject & Sign
- Claim
- Release
- Dispute
- Executed Guarantee

29. THE FINAL USER EXPERIENCE

The user should experience it as:

SUPPLIER PORTAL





30. Product Differentiator

The platform is therefore not simply:

“A system that issues guarantees.”

It becomes:

A complete digitally signed guarantee lifecycle platform

covering:

Request → Signature → Bank Processing → Approval → Guarantee Issuance → Official PDF → Secure Delivery → Verification → Beneficiary Acceptance → Digital Signature → Active Guarantee → Claim → Release → Dispute → Closure.